



Mark Scheme (Results)

June 2024

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1B)
Unit 1: Principles and Applications of Science I
(Biology)

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Unit 1: Principles and Applications of Science I (Biology)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

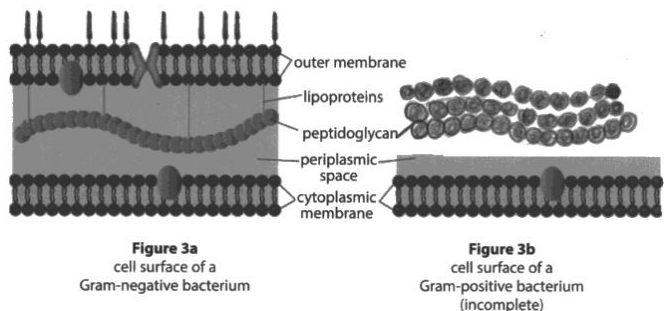
The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)	(light) microscope	accept electron microscope ignore magnifying glass ignore lenses	1
1(b)	Alveoli	accept air sac	1
1(c)(i)	D smoking cigarettes		1
1(c)(ii)	Any one from: (squamous epithelium) thickens / inflamed (1) decreases elasticity (1)	 Ignore references to surface area	1
1(d)	<u>conversion (1)</u> (4.92 x 1000 =) 4920 (µm) <u>substitution (1)</u> 4920 ÷ 82 <u>evaluation (1)</u> 60 (X) OR <u>conversion (1)</u> (82 ÷ 1000 =) 0.082 (µm) <u>substitution (1)</u> 4.92 ÷ 0.082 <u>evaluation (1)</u> 60 (X)	60 alone gains 3 marks	3
Total = 7 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	Cells that have the ability to develop into many different cell types are called stem (1) cells. Cells become specialised for a particular function by a process known as differentiation . (1)	accept undifferentiated / pluripotent / totipotent ignore unspecialised	2
2(b)(i)	Any one from: contains nutrients transcription factors (medium for) chemical reactions	accept named examples, e.g. protein accept named example	1
2(b)(ii)	(sperm cell) needs <u>large</u> amount of energy for {movement / swimming / propulsion} (1) mitochondria {release energy/ produce ATP} by (aerobic) <u>respiration</u> (1)	reject "produce energy"	2
2(b)(iii)	<u>conversion (1)</u> (0.1 x 1000 =) 100 (µm) <u>evaluation (1)</u> (100 ÷ 4 =) 25 (times larger than the sperm cell head) OR <u>conversion (1)</u> (4 ÷ 1000 =) 0.004 (µm) <u>evaluation (1)</u> (0.1 ÷ 0.004 =) 25 (times larger than the sperm cell head)	25 alone gains 2 marks	2
Total = 7 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)	peptidoglycan layer only added (1) thicker than Gram-negative peptidoglycan (1) labelled as peptidoglycan (1)  Figure 3a cell surface of a Gram-negative bacterium Figure 3b cell surface of a Gram-positive bacterium (incomplete)	reject any outer membrane drawn and lipoproteins	3
3(b)	enables the bacterium to be classified as Gram-positive or Gram-negative (1) so that appropriate antibiotic/treatment can be used (1)	accept to see if a named antibiotic will work, if it only works on one type of Gram bacteria	2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	C excitatory, inhibitory		1
4(b)	Any four from: atropine is a (competitive) <u>antagonist</u> (1) (atropine) has similar shape to acetylcholine (1) (atropine's) shape is complementary (to the receptor) (1) (atropine) binds to the receptors (1) (atropine) prevents/ blocks acetylcholine binding (1) reduces the effect in the post-synaptic membrane (1)	accept a description of complementary	4
Total = 5 marks			

Question Number	Indicative content
5	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> muscle fibres are made of smaller fibres / myofibrils myofibrils consist of sarcomeres sarcomeres are the part of the muscle that becomes shorter a nerve impulse arrives at a neuromuscular junction and causes the release of acetylcholine / neurotransmitter this causes an action potential/ depolarisation of the muscle sarcolemma sarcoplasmic reticulum is stimulated to release calcium ions calcium ions bind to troponin troponin/tropomyosin displace off actin exposing myosin binding sites myosin heads already have ATP attached and now energy is gained from ATP hydrolysis because myosin binding sites act as ATPase cross bridges form (between myosin heads and actin) cross bridges on myosin pull on actin power stroke actin filaments slide past myosin to middle of sarcomere I band shortens A band stays the same M band stays the same Z lines pulled closer together muscle shortens and contracts cross bridge detaches from binding sites on actin using ATP muscle fibre then relaxes and lengthens calcium ions actively pumped back into sarcoplasmic reticulum

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts / concepts with generalised comments made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts / concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.



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